Contents

The signal model — the algebra of the chain **1**
1. The transmitter — `s(t)` . . . 1
2. Propagation — one source to one receiver . . . 2
   2.1 Emission time (the light-time solve) . . . 2
   2.2 Earth rotation (Sagnac) . . . 2
   2.3 Amplitude — free-space path loss + antenna . . . 3
   2.4 The received phase . . . 3
3. Superposition — the incident field . . . 4
4. The receiver front-end — operators on `r(t)` . . . 4
   4.1 Antenna (per source) — `G_ant` . . . 4
   4.2 The oscillator (per source, via `δ_o`) — `θ_{osc}(t)` . . . 4
   4.3 RF/IF gain + AGC (post-sum) — `g` . . . 5
   4.4 Nonlinearity (post-sum) — IP3 . . . 5
   4.5 Thermal noise (post-sum) — `n(t)` . . . 5
   4.6 The ADC — **the collapse** . . . 5
5. The complete forward model . . . 5
6. The inverse — what an estimator does . . . 6
7. Notable exclusions — what the model does *not* capture . . . 6
8. Glossary . . . 7
9. Equation ↔ code map . . . 7
10. References & relation to prior art . . . 8

The signal model — the algebra of the chain

This is the mathematical specification the code implements. The guiding principle: **everything is a continuous function of time; the only collapse to discrete samples is the ADC.** The transmitter is a function `s(t)`; propagation, the oscillator, the antenna and the front-end are *operators* on it; sampling + quantisation is the single discrete, lossy step. Because every operator is exact and stateless-in-form, an *estimator* can run the same algebra **backwards** (§6).

The numbered equations **(E1)–(E10)** each cite the code that implements them; §9 collects the equation ↔ code map. Symbols are in the glossary (§8). What the model deliberately leaves out is in **§7 (notable exclusions)** — read it before trusting a result that depends on an unmodelled effect.

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1. The transmitter — `s(t)`

A transmitter is a **stateless complex-baseband function** of emission time (`src/signal/mod.rs::SignalModel`), riding a nominal RF carrier `f_tx`:

$$s_{\mathrm{RF}}(t) \;=\; \Re\Big[\, s(t)\, e^{\,j2\pi f_{tx} t} \,\Big], \qquad s(t) \in \mathbb{C}.$$

`s(t)` is the *information*; `f_tx` is the carrier. Concrete models define `s(t)`:

signal	`s(t)`
CW tone (offset `f_b`)	$s(t) = e^{\,j2\pi f_b t}$
BPSK (bits `b_k`, rate `R`)	$s(t) = b_{\lfloor Rt \rfloor} \in \{\pm 1\}$

signal	`s(t)`
GPS C/A (PRN, NAV `d`)	$s(t) = c_{\mathrm{PRN}}(t)\, d(t) \in \{\pm 1\}$

The crucial property: **`s` is never pre-sampled.** Doppler, delay and clock all act by changing *where* `s` is evaluated, not by resampling a buffer — which is what keeps the model exact.

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2. Propagation — one source to one receiver

2.1 Emission time (the light-time solve)

The receiver timestamps each sample with its **own clock**, which reads `t = n/f_s` at sample `n`. That clock is imperfect: its error `δ_o(t)` (§4.2) means the **true** (coordinate) time at which the sample is actually captured is

$$t_{\mathrm{utc}} \;=\; t - \delta_o(t).$$

All geometry is evaluated at true time. A sample whose true reception instant is `t_utc` was emitted earlier, at the **emission time** `τ`, because the signal travelled the slant range at `c`. With the source at `P_s` and the receiver at `P_o`, `τ` solves the implicit equation:

$$\boxed{\;\tau \;+\; \frac{\left\|\mathbf{P}_s(\tau) - \mathbf{P}_o(t_{\mathrm{utc}})\right\|}{c} \;=\; t_{\mathrm{utc}} \;=\; t - \delta_o(t)\;} \tag{E1}$$

> **code** — the pipeline forms the true-time grid `t_utc_rel = t_local_rel − delta_t` (`pipeline.rs`), then `channel::solve_emission_times_for` runs the per-sample loop.

i.e. emission time + light-travel = true reception time. **This is where the clock enters the geometry**: `τ` is anchored to `t_utc = t − δ_o`, so every quantity derived from it — the envelope `s(τ)`, the range `r`, the Doppler — carries the clock. The sample is *indexed* by clock time `t`; the physics happens at `t_utc`. Everything dynamic — Doppler, code-rate Doppler, sample-rate skew — falls out of `τ` varying across samples, with no separate "apply Doppler" or "apply clock" step. This is the exact (*wideband*) treatment of Doppler as a time-scaling of the waveform rather than the narrowband `e^{j2πf_d t}` frequency-shift approximation [KW65; VanTrees-III; Bello63]; in GNSS terms the carrier and code Doppler are then automatically the same `ṙ/c` scaling [KH17]. The light-time solve itself is the standard ephemeris "reception-case" construction [SPICE96; IERS2010].

2.2 Earth rotation (Sagnac)

`P_s(τ)` and `P_o(t_utc)` are ECEF positions at *different* instants, but the geometric path must be measured in one frame. The source is rotated into the reception-frame by Earth's rotation over the transit, `Δ = t_utc − τ = r/c` (linearised, exact to sub-mm):

$$\tilde{\mathbf{P}}_s = R_3(\omega_\oplus\, \Delta)\, \mathbf{P}_s(\tau) \;\approx\; \big(P_{s,x} + \omega_\oplus \Delta\, P_{s,y},\;\; P_{s,y} - \omega_\oplus \Delta\, P_{s,x},\;\; P_{s,z}\big). \tag{E2}$$

> **code** — the `wtau`/`sx_rot`/`sy_rot` block inside `channel::solve_emission_times_for`.

The range used everywhere below is this Sagnac-corrected slant range, $r(t) = \|\tilde{\mathbf{P}}_s - \mathbf{P}_o\|$. Omitting it costs ~tens of ns — significant for a timing reference [IERS2010]. (The OD side, `common.emission_ranges`, solves the *same* equation — reception- vs emission-time geometry was the light-time bug that cost ~50 m of along-track until fixed.)

2.3 Amplitude — free-space path loss + antenna

$$A(t) \;=\; \underbrace{\frac{\lambda}{4\pi\, r(t)}\sqrt{P_{tx}}}_{\text{FSPL (volts)}} \cdot \underbrace{\sqrt{G_{\text{ant}}(\theta_s(t))}}_{\text{rx antenna}}, \qquad \lambda = \frac{c}{f_{tx}}. \tag{E3}$$

> **code** — `amp_factor`/`amplitude` (FSPL) × `geom_gain` (antenna √G), in `channel::accumulate_transmitter_contribution`'s mix loop.

`P_tx` is EIRP (W); the `λ/4πr` factor is the free-space field amplitude; `√` because amplitude, not power. `G_ant` is the receiver antenna's power gain at the source's arrival elevation `θ_s(t)` (§4.1). lipika amplitudes are in **√W**, so `|·|²` is power directly.

2.4 The received phase

Two phase processes meet at the mixer: the **arriving carrier**, whose phase was set at the source at emission time `τ`, and the receiver **LO**. The LO is generated from the same oscillator that defines the clock, so at sample `n` its phase is exactly `2πf_lo·t` — clock time, *by definition of the clock*. The baseband phase is the arriving carrier phase minus the LO phase:

$$\boxed{\varphi \;=\; \underbrace{\arg s(\tau)}_{\text{message}} + \underbrace{2\pi f_{tx}\,\tau}_{\text{arriving carrier}} - \underbrace{2\pi f_{lo}\,t}_{\text{LO (clock time)}}} \tag{E4}$$

Only two frequencies appear: `f_tx` rides on `τ` (true-time geometry), `f_lo` rides on `t` (clock time). Expanding with `τ = t_utc − r/c = t − δ_o − r/c` separates the familiar terms and matches the code exactly:

$$\varphi \;=\; \arg s(\tau) \;-\; 2\pi f_{tx}\Big(\tfrac{r}{c} + \delta_o\Big) \;+\; 2\pi(f_{tx} - f_{lo})\, t.$$

> **code** — the `phase = neg_tau_src*(delay+delta_t) + tau_if*t_local` line in `channel::accumulate_transmitter_contribution` (`neg_tau_src=−2πf_tx`, `tau_if=2π(f_tx−f_lo)`, `delta_t=δ_o`, `t_local=t`). The envelope `s(τ)` is evaluated at `τ` from the true-time solve (E1), so it carries `δ_o` too.

Reading the terms:

- **message** `arg s(τ)` — sampled at the emission time `τ`. Because `τ` is anchored to `t_utc = t − δ_o` (E1), the clock shifts the **code phase** here.
- **carrier delay** `−2πf_tx·r/c` — carrier phase over the light path; its time-derivative *is* the carrier Doppler, emerging automatically as `τ` walks.
- **clock** `−2πf_tx·δ_o` — the receiver oscillator, via the LO time base. It carries the **same sign and the same `δ_o`** as the code-phase shift in `arg s(τ)`, so the clock acts as a *common time bias* on code and carrier alike: the apparent delay is `t − τ = r/c + δ_o`, i.e. **pseudorange `= r + c·δ_o`** (the standard GNSS observation model [ME06; KH17]). This is why clock and LO are one object — one `δ_o` drives sample timing and carrier together.
- **IF / de-ramp** `2π(f_tx−f_lo)t` — for an off-tune LO the residual sits at the intermediate frequency `f_tx−f_lo`; vanishes on-tune (`f_lo=f_tx`).

(The `−2πf_tx·δ_o` clock term paired with the clock-time de-ramp is algebraically identical to the `−2πf_lo·δ_o` "LO phase-noise" form with the de-ramp expressed in true time `t_utc`; they differ only in bookkeeping. The code keeps the de-ramp on clock time, hence `f_tx`.)

For an on-tune receiver (`f_lo=f_tx`) this collapses to the form in the code:

$$\varphi = \arg s(\tau) \;-\; 2\pi f_{tx}\Big(\tfrac{r(t)}{c} + \delta_o(t)\Big), \qquad \tau = t - \delta_o - \tfrac{r}{c}.$$

So one source contributes the continuous complex sample $A\,|s(\tau)|\,e^{j\varphi}$.

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3. Superposition — the incident field

A receiver band sees the **sum over all in-band sources** (the `bands_overlap` gate decides "in-band"):

$$\boxed{r(t) \;=\; \sum_{i \in \text{sources}} A_i(t)\, s_i(\tau_i(t))\, e^{j\varphi_i(t)}} \tag{E5}$$

> **code** — `pipeline::generate_band` sums each transmitter's `accumulate_transmitter_contribution` into the band buffer `acc` (fixed order ⇒ deterministic). `bands_overlap` gates the source set.

This is still an exact continuous function. Horizon occlusion enters here: a source below the receiver's local horizon is dropped (`A_i \to 0`).

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4. The receiver front-end — operators on `r(t)`

The front-end is a composition of operators (`src/rx`). Some act **per source** (woven into §2, because they depend on each source's geometry); the rest act on the **summed band** `r(t)`. The chain, in order:

4.1 Antenna (per source) — `G_ant`

A gain on the arrival direction. The chosen self-phased QFH is azimuthally symmetric, so gain depends on elevation `θ` only:

$$G_{\text{ant}}(\theta)\;[\text{dBi}] = \begin{cases} G_{\text{horiz}} + (G_{\text{peak}} - G_{\text{horiz}})\,\theta/90^\circ & \theta \ge 0 \\ G_{\text{horiz}} - \text{F/B} & \theta < 0 \text{ (back-lobe)} \end{cases} \tag{E6}$$

applied as `√G` to each source's amplitude (§2.3). `θ` is the same arrival angle the horizon mask uses.

> **code** — `rx::Antenna::amplitude_at`; folded into the `geom_gain` pass of `channel::accumulate_transmitter_contribution`.

4.2 The oscillator (per source, via `δ_o`) — `θ_{osc}(t)`

The node has **one** oscillator phase process. Its fractional-frequency noise `y(t)` integrates to the **time error** `δ_o(t)` that appears in `φ` (§2.4):

$$\delta_o(t) \;=\; \underbrace{c_0}_{\text{offset}} + \underbrace{m\,t}_{\text{drift}} + \underbrace{\varepsilon_{\text{wFM}}(t)}_{\text{white FM (Allan } \sigma_1\text{)}} + \underbrace{\varepsilon_{\text{fFM}}(t)}_{\text{flicker FM (Allan floor)}} \tag{E7}$$

> **code** — `clock::NodeClockSpec` (the parameters), `clock::ClockStream` (the streaming `δ_o(t)`: deterministic part + white-FM Wiener walk), `clock::Flicker` (the flicker-FM floor).

- `c₀` — static bias (per-node, unobservable absolute; a nuisance in the OD).
- `m` — static fractional-frequency offset (ppm-scale drift).
- `ε_wFM` — white-FM **Wiener walk**, calibrated so $\sigma_y(\tau{=}1\text{s}) = \sigma_1$; Allan $\propto \tau^{-1/2}$.

- `ε_fFM` — **flicker FM** (1/f), a flat Allan **floor** — the oscillator-grade knob (TCXO ≈ 1e-9…1e-10, OCXO ≈ 1e-12…1e-13). Built as a sum of log-spaced AR(1) relaxations, self-calibrated (`clock::Flicker`).

The `c₀ + m·t +` power-law decomposition is the standard frequency-stability model [Barnes71; IEEE1139]; the white-/flicker-FM synthesis (Allan-correct, causal, scale-invariant) follows the discrete power-law-noise method of [Kasdin92; Kasdin95].

The **same `δ_o`** sets the sample-timing grid and the LO time base. So **clock = LO = sample clock**: one oscillator drives the sample timing — hence `τ`, the code phase and range, via `t_utc = t − δ_o` (§2.1) — **and** the carrier phase, via the `−2πf_tx·δ_o` term (§2.4). Because it is one `δ_o`, code and carrier see the *same* clock: a common time bias, pseudorange `= r + c·δ_o`.

4.3 RF/IF gain + AGC (post-sum) — `g`

A linear scalar (dB) lifting the weak (FSPL-attenuated) signal into the ADC range: $y \mapsto g\,y$.

4.4 Nonlinearity (post-sum) — IP3

Memoryless third-order LNA/mixer model with input intercept power `P_{\text{IP3}}`:

$$y \;\mapsto\; y\Big(1 - \tfrac{4}{3}\,\frac{|y|^2}{P_{\text{IP3}}}\Big). \tag{E8}$$

code — `rx::PostOp::Nonlinearity`.

Small-signal linear; gain-compresses and produces IM3 at strong input. The binding spec for terrestrial rejection. (`P_IP3` from the dBm datasheet value, direct since amplitude² is power.) This memoryless-cubic intercept model and the gain/NF/quantise front-end chain are the standard behavioural receiver model [Betz16; Borre07].

4.5 Thermal noise (post-sum) — `n(t)`

Additive complex AWGN at the receiver floor $P_n = k_B T B$ (`B` = sample rate): $y \mapsto y + n,\ n \sim \mathcal{CN}(0, P_n)$.

4.6 The ADC — the collapse

The only discrete, lossy, non-invertible step: sample on the **clock** grid `t_n=n/f_s` (§2.1 — `t_n` is clock time; the geometry inside `φ, s(τ), A` is evaluated at the true time `t_n − δ_o`, and aperture jitter is a further time-warp of this grid), clip to the full scale `±V_{fs}`, and quantise to `b` bits:

$$x[n] \;=\; Q_b\big(\text{clip}_{\pm V_{fs}}(y(t_n))\big), \qquad Q_b(v) = \Delta\,\text{round}(v/\Delta),\ \ \Delta = \frac{V_{fs}}{2^{b-1}-1}. \tag{E9}$$

code — `rx::PostOp::Adc`.

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5. The complete forward model

Putting it together — what lipika computes for output sample `n` of a band:

$$\boxed{x[n] \;=\; Q_b\Bigg(\text{clip}\Bigg[\,\mathcal{N}\Big(g\sum_i A_i(t_n)\,s_i(\tau_i)\,e^{j\varphi_i(t_n)}\Big) \;+\; n[n]\,\Bigg]\Bigg)} \quad t_n = \tfrac{n}{f_s} \tag{E10}$$

code — `pipeline::generate_band` per band: the `Σ` (E5) + `add_awgn_at_power` + `RxChain::apply_post_sum` (gain, NL, ADC), then `sink.push`.

Everything inside `Q_b(clip(·))` is an exact continuous expression evaluated at `t_n`; `Q_b` is the single collapse. With an unconfigured front-end (`g=1`, no NL, ideal ADC) this reduces to the plain `Σ` — which is why adding the front-end leaves old scenes byte-identical.

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6. The inverse — what an estimator does

Because every operator above is exact and the lossy ones (NL, thermal, quantisation) are well-characterised noise, an estimator can run the **same algebra backwards** on `x[n]`, inverting the *invertible* operators and treating the rest as noise:

- **De-rotation** = inverse of §2.4's carrier+IF terms: multiply by `e^{-j[2π(f_tx-f_lo)t - 2πf_tx r̂/c]}` using a *predicted* range `r̂` (from ephemeris+position, or the orbit fit) → leaves the slow residual `-2πf·δ_o + noise`.
- **Code-phase prediction** = inverse of the §2.1 time-warp: predict `τ` from the broadcast ephemeris + surveyed position → the measured-minus-predicted code shift is `δ_o` (the forward model put it there via `t_utc = t − δ_o`, so it is recoverable).
- **Clock recovery** = estimating `δ_o(t)` (E7): since the clock is a common time bias (pseudorange `= r + c·δ_o`, §2.4) it is common-mode across all visible SVs (geometry wiped); code resolves integer-ns, carrier carries the ps fraction.

Because forward and inverse are the *same* `φ`, `r`, `δ`, a simulator and an estimator built on this algebra cannot drift apart — and a result proved against the forward geometry (e.g. the light-time term, E1–E2) transfers directly to the inverse. This analysis-by-synthesis stance — invert the forward model directly rather than track loosely — is the basis of GNSS direct position estimation and vector tracking [Closas07]. **Caveat:** this is *internal* consistency, not fidelity. Any effect the forward model omits (§7) is invisible to this round-trip, so passing it proves the algebra, not the physics.

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7. Notable exclusions — what the model does *not* capture

The channel is **vacuum truth**: geometry, free-space loss, antenna, one *receiver* clock, and a characterised front-end. The effects below are real but deliberately **not** modelled today. They are listed so an estimator knows what its "truth" omits, and so the self-consistency of §6 is not mistaken for real-world accuracy.

- **Ionospheric propagation.** No group delay / phase advance / TEC. At UHF this is the **largest** missing term — tens to >100 ns of group delay for typical TEC, i.e. *larger* than the Sagnac correction the geometry does include (it scales as `1/f²`, §2.2). The network's TEC / atmospheric-sounding ambitions live exactly here; today the ionosphere is an *estimand*, not a simulated channel.
- **Tropospheric delay.** No neutral-atmosphere delay — small versus the ionosphere at these frequencies, but nonzero and growing toward the horizon.
- **Transmitter clock.** `s(t)` is phase-perfect: the *source* oscillator is ideal (no transmit `δ_o`, no transmit phase noise). A per-source clock is a clean extension (a mirror of §4.2), but for downlink / TDOA we currently assume emitters are stable. Adding the term doesn't hurt; we just hold it at zero.
- **Polarisation.** No polarisation state or mismatch — the QFH is a scalar gain `G_ant(θ)` (§4.1). No circular/linear mismatch loss, no Faraday rotation.

- **Receiver noise figure.** Thermal noise is `k_B T B` at the floor (§4.5) with no noise-figure / system-temperature budget, and is added *post-gain* — so absolute SNR tracks the configured gain `g` rather than a cascade NF.
- **Relativistic terms.** First-order (classical) Doppler only, via the light-time walk of `τ`; second-order time dilation and gravitational frequency shift are dropped (`~6e-10` fractional — borderline only for ps-class carrier-phase work).
- **Multipath / scattering.** One direct ray per source; no ground bounce, no local scatterers.

None of these conflict with the "evaluate the operator, don't resample" discipline; they are scoped out, not overlooked. When one is added it slots in as another exact operator on `s(τ)` or another term in `δ`.

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8. Glossary

symbol	meaning
`t`	node-**clock** time of a sample (the timestamp `n/f_s`; the LO time base)
`t_utc`	true (coordinate) reception time `= t − δ_o`; geometry is solved here
`τ`	emission time (light-time solve at `t_utc`, §2.1)
`s(t)`	transmitter complex baseband (§1)
`f_tx`, `f_lo`	source RF carrier; receiver band LO
`f_s`, `B`	band sample rate (= noise bandwidth)
`P_s, P_o`	source / observer ECEF position
`r(t)`	Sagnac-corrected slant range
`c`, `λ`, `ω⊕`	speed of light; wavelength `c/f_tx`; Earth rotation rate
`P_tx`	transmit EIRP (W); lipika amplitude^2 is power
`G_ant(θ)`	rx antenna power gain at elevation `θ`
`δ_o(t)`	node clock/oscillator time error (§4.2)
`c₀, m`	clock static offset; fractional-frequency drift
`σ_1`	white-FM Allan deviation at τ=1 s
`ε_wFM, ε_fFM`	white-FM walk; flicker-FM floor
`g`, `P_IP3`	RF/IF gain; input third-order intercept (W)
`P_n=k_BTB`	thermal noise power
`V_fs`, `b`, `Δ`	ADC full scale; bits; LSB
`Q_b`, `n` (index)	`b`-bit quantiser; output sample index

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9. Equation ↔ code map

Each formula and the function that implements it. Source files are under `src/`; every function below carries a `(E…)` back-reference in its doc-comment.

eq	what	code
E1	emission-time (light-time) solve	`channel::solve_emission_times_for`

eq	what	code
E2	Sagnac (Earth-rotation) correction	same loop (`wtau`/`sx_rot`/`sy_rot`)
E3	amplitude — FSPL × antenna √G	`channel::accumulate_transmitter_contribution` (`amp_factor`, `geom_gain`)
E4	received phase `φ(t)`	`channel::accumulate_transmitter_contribution` (the `phase = …` line)
E5	superposition `r(t)`	`pipeline::generate_band` (sum of contributions)
E6	antenna gain `G_ant(θ)`	`rx::Antenna::amplitude_at`
E7	oscillator `δ_o(t)`	`clock::{NodeClockSpec, ClockStream, Flicker}`
E8	IP3 nonlinearity	`rx::PostOp::Nonlinearity`
E9	ADC collapse (clip + quantise)	`rx::PostOp::Adc`
E10	full forward `x[n]`	`pipeline::generate_band` (Σ + noise + `RxChain::apply_post_sum`)

The reverse links live in the code: each listed function's doc-comment names its equation, so `grep -rn "SIGNAL_MODEL" src/` walks from code back to the maths.

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10. References & relation to prior art

No part of this model is novel physics. Each ingredient is standard; the contribution is the *integration* — exact wideband Doppler, a behavioural front end, a physical clock-noise process, and an explicit forward/inverse symmetry, kept as one continuous operator algebra whose only discrete step is the ADC. The literature each piece rests on:

- **Doppler as time-warp / exact (wideband) channel** (§1, §2.1, §2.4)
  - **[Bello63]** P. A. Bello, "Characterization of randomly time-variant linear channels," *IEEE Trans. Communications Systems*, 11(4):360–393, 1963.
  - **[KW65]** E. J. Kelly and R. P. Wishner, "Matched-filter theory for high-velocity, accelerating targets," *IEEE Trans. Military Electronics*, MIL-9(1):56–69, 1965.
  - **[VanTrees-III]** H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part III: Radar–Sonar Processing*, Wiley, 1971 (repr. 2001) — wideband ambiguity, Doppler as time-scaling.
- **Light-time solve & Earth-rotation (Sagnac)** (§2.1–2.2)
  - **[SPICE96]** C. H. Acton, "Ancillary data services of NASA's Navigation and Ancillary Information Facility (NAIF)," *Planetary and Space Science*, 44(1):65–70, 1996 — the ephemeris "reception-case" light-time construction.
  - **[IERS2010]** G. Petit and B. Luzum (eds.), *IERS Conventions (2010)*, IERS Technical Note No. 36 — Earth-rotation/Sagnac and relativistic propagation terms.
- **GNSS observation model & code/carrier Doppler** (§2.4, §4.2)
  - **[ME06]** P. Misra and P. Enge, *Global Positioning System: Signals, Measurements, and Performance*, 2nd ed., Ganga-Jamuna Press, 2006 — pseudorange `ρ = r + c·δt`.
  - **[KH17]** E. D. Kaplan and C. J. Hegarty (eds.), *Understanding GPS/GNSS: Principles and Applications*, 3rd ed., Artech House, 2017 — carrier and code Doppler as the shared `ṙ/c` scaling.
- **Clock / oscillator noise** (§4.2)
  - **[Barnes71]** J. A. Barnes et al., "Characterization of frequency stability," *IEEE Trans. Instrumentation and Measurement*, IM-20(2):105–120, 1971.

  - **[IEEE1139]** IEEE Std 1139, *Definitions of Physical Quantities for Fundamental Frequency and Time Metrology — Random Instabilities*.
  - **[Kasdin92]** N. J. Kasdin and T. Walter, “Discrete simulation of power law noise,” *Proc. IEEE Frequency Control Symposium*, pp. 274–283, 1992.
  - **[Kasdin95]** N. J. Kasdin, “Discrete simulation of colored noise and stochastic processes and 1/f^α power law noise generation,” *Proc. IEEE*, 83(5):802–827, 1995.
- **Receiver front-end (behavioural)** (§4.3–4.6)
  - **[Betz16]** J. W. Betz, *Engineering Satellite-Based Navigation and Timing*, IEEE Press/Wiley, 2016 — noise figure, nonlinearity, quantisation effects.
  - **[Borre07]** K. Borre, D. M. Akos, et al., *A Software-Defined GPS and Galileo Receiver*, Birkhäuser, 2007 — IF-sampling front end and quantisation loss.
- **Forward-model inversion (analysis by synthesis)** (§6)
  - **[Closas07]** P. Closas, C. Fernández-Prades, J. A. Fernández-Rubio, “Maximum likelihood estimation of position in GNSS,” *IEEE Signal Processing Letters*, 14(5):359–362, 2007 — direct position estimation; the forward model inverted directly.